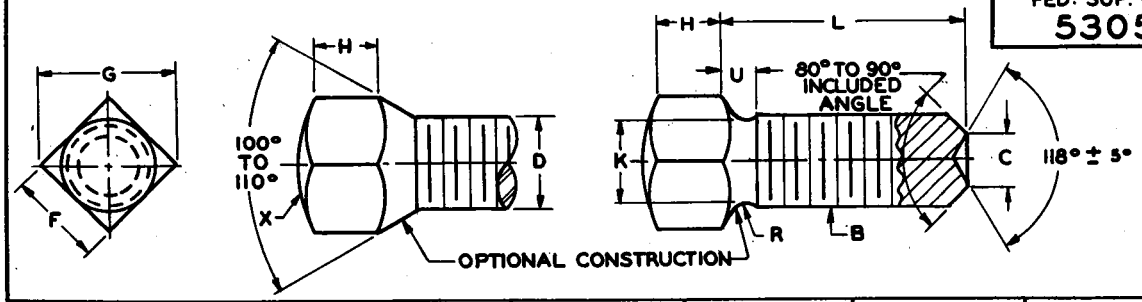
FED. SUP. CLASS.
5305



D	Nominal Diameter	1/4	5/16	3/8	7/16	1/2
B	Threads Per Inch	20UNC-2A	18UNC-2A	16UNC-2A	14UNC-2A	13UNC-2A
K	Diameter of Neck Relief Max	.185	.240	.294	.345	.400
	Min	.170	.225	.279	.330	.385
F	Width Across Flats Max	.250	.3125	.375	.4375	.500
	Min	.240	.302	.362	.423	.484
G	Width Across Corners Min	.331	.415	.497	.581	.665
H	Height of Head Nom	.187	.234	.281	.328	.375
	Max	.196	.245	.293	.341	.389
	Min	.178	.224	.270	.315	.361
R	Radius of Neck Relief Max	.032	.036	.041	.046	.050
U	Width of Neck Relief Max	.100	.111	.125	.143	.154
X	Radius of Head	.625	.781	.937	1.094	1.250
C	Diameter of Point Max	.132	.172	.212	.252	.291
	Min	.118	.156	.194	.232	.270

L Length	Tolerance	Dash No.	Fin	Dash No.	Fin	Dash No.	Fin	Dash No.	Fin	Dash No.	Fin
3/8	+0 -1/32	1		16							
1/2		2		17							
5/8		3		18		34		51			
3/4		4		19		35		52		68	
7/8		5		20		36		53		69	
1	+0 -1/16	6		21		37		54		70	
1-1/4		7		22		38		55		71	
1-1/2		8		23		39		56		72	
1-3/4		9		24		40		57		73	
2		10		25		41		58		74	
2-1/4	+0 -3/32			26		42		59		75	
2-1/2				27		43		60		76	
3				28		44		61		77	
3-1/2						45		62		78	
4										79	
										80	

MATERIAL: STEEL, CARBON, IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.
HARDNESS: SETSCREWS SHALL BE HARDENED BY HEAT TREATMENT TO A HARDNESS OF ROCKWELL C30 TO 37 OR SHALL BE SURFACE HARDENED TO NOT LESS THAN ROCKWELL 30N64.
PROTECTIVE COATING: CADMIUM PLATED, IN ACCORDANCE WITH SPECIFICATION QQ-P-416, TYPE II, CLASS 3.
ZINC PLATED, IN ACCORDANCE WITH SPECIFICATION QQ-Z-325, TYPE II, CLASS 3.
THREADS: THE THREADS SHALL BE IN ACCORDANCE WITH SCREW-THREAD STANDARDS FOR FEDERAL SERVICES HANDBOOK H-28.
DIMENSIONS: ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.
PART NUMBER: THE MS PART NUMBER CONSISTS OF THE MS NUMBER, PLUS THE DASH NUMBER. EXAMPLE: MS51054-1.
NOTES: (1) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.
(2) THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.

THIS DOCUMENT HAS BEEN PROMULGATED BY THE DEPARTMENT OF DEFENSE AS THE MILITARY STANDARD TO LIMIT THE SELECTION OF THE ITEM, PRODUCT, OR DESIGN COVERED HEREIN IN ENGINEERING, DESIGN, AND PROCUREMENT. THIS STANDARD SHALL BECOME EFFECTIVE NOT LATER THAN 90 DAYS AFTER THE LATEST DATE OF APPROVAL SHOWN.

APPROVED 7 MAY 1958 REVISED

CUSTODIANS	OTHER INT.
A - ORD	A-CESIGT
N - SHIPS	N-AMCOR
AF - USAF	AF-

MILITARY STANDARD

SETSCREW, SQUARE HEAD, CUP POINT, CARBON STEEL, CADMIUM OR ZINC, UNC-2A

MS51054

PROCUREMENT SPECIFICATION: FF-S-103
SUPERSEDES:
SHEET 1 OF 2

DD FORM 1 OCT 52 672-1

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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FED. SUP. CLASS.
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		5/8	3/4	7/8	1	
D	NOMINAL DIAMETER	5/8	3/4	7/8	1	
B	THREADS PER INCH	11UNC-2A	10UNC-2A	9UNC-2A	8UNC-2A	
K	DIAMETER OF NECK RELIEF MAX	.507	.620	.731	.838	
	MIN	.492	.605	.716	.823	
F	WIDTH ACROSS FLATS MAX	.625	.750	.875	1.000	
	MIN	.606	.729	.852	.974	
G	WIDTH ACROSS CORNERS MIN	.833	1.001	1.170	1.337	
H	HEIGHT OF HEAD NOM	.468	.562	.656	.750	
	MAX	.485	.582	.678	.774	
	MIN	.452	.544	.635	.726	
R	RADIUS OF NECK RELIEF MAX	.059	.065	.072	.081	
U	WIDTH OF NECK RELIEF MAX	.182	.200	.222	.250	
X	RADIUS OF HEAD	1.562	1.875	.2187	2.500	
C	DIAMETER OF POINT MAX	.371	.450	.530	.609	
	MIN	.347	.425	.502	.579	

L LENGTH	TOLERANCE	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN
3/4	+0	86									
7/8	-1/32	87									
1		88		104							
1-1/4	+0	89		105							
1-1/2	-1/16	90		106		120					
1-3/4		91		107		121		136			
2		92		108		122		137			
2-1/4		93		109		123		138			
2-1/2		94		110		124		139			
3	+0	95		111		125		140			
3-1/2	-3/32	96		112		126		141			
4		97		113		127		142			
4-1/2		98		114		128		143			
5						129		144			
						130		145			

NOTE: FOR MATERIAL, PROTECTIVE COATING AND OTHER PERTINENT DATA, SEE SHEET 1.

APPROVED 7 MAY 1958 REVISED

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIANS: A - ORD; N - SHIPS; AF - USAF
OTHER INT.: A- CESIGT; N- AMCOR; AF-

MILITARY STANDARD

SETSCREW, SQUARE HEAD, CUP POINT, CARBON STEEL, CADMIUM OR ZINC, UNC-2A

MS51054

PROCUREMENT SPECIFICATION: FF-S-103

SUPERSEDES:

SHEET 2 OF 2

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